# Question and design

Does model width improve contact accuracy at the same one-epoch exposure? Baseline: exp277, native plus MPNN corpus, 24 layers. Hidden width 2048 -> 4096; MLP width 8192 -> 16384. Keep 32 query / 8 KV heads; head width 64 -> 128. Keep data, optimizer schedule, seeds and global batch fixed.

# Architecture and smoke verified

5,861,984,256 parameters, versus exp277's 1,471,374,336 (3.984x). Architecture, tokenizer, data and recovery checks passed. 64-H100 smoke: ten updates, train loss 8.56948, eval loss 8.03575. 2.821 seconds/update; native and HF checkpoints saved with tokenizer. Compressed smoke schedule was spiky; this is an execution check.

# Current status — October 19

GPU production running: step 48,451, 18.19% of one epoch. Train loss 2.893; validation loss 3.123, previously 3.465. Complete rolling checkpoint step 48,452; permanent step 26,634. 32 H100s: 5.61 s/update; about 14.2 compute days remaining. Queueing, validation, checkpoints and preemptions add wall time.

# TPU trial succeeded

Interactive v5p-64: 32 chips across 8 VMs; all 20 updates completed. Finished October 16; no worker failures or preemptions. Native and HF step-19 checkpoints verified; tokenizer included. Median steps 5–19: 7.40 s/update, about 32% longer than GPU steps. Short native-only smoke; throughput is preliminary, not cost-normalized. Production continues on GPUs. No bulk transfer or migration performed.

# Matched validation: no width benefit yet

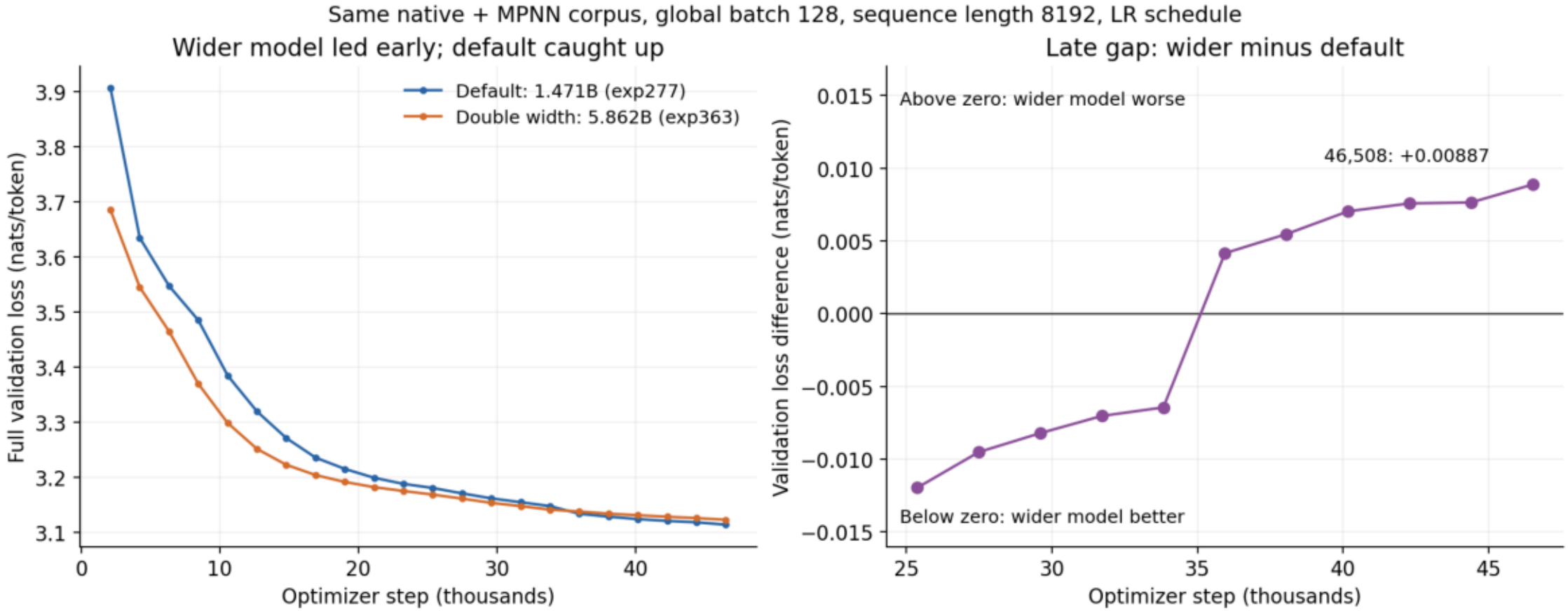
Same native + MPNN corpus, global batch, validation and optimizer schedule. Wider model led early; default overtook it between steps 33,824 and 35,938. Default has lower loss at all six subsequent full evaluations. At step 46,508: default 3.11447; wider 3.12334. Gap +0.00887 nats/token, or 0.89% higher perplexity. Single seed, about 18% of the epoch; baseline LR retained without retuning. Both losses keep falling. No contact-accuracy evaluation for the wider model yet.

# Other architectures: context, not a common leaderboard

exp108 depth-scaled 2.93B: all three LR trials unstable around 6.8k updates. exp108 width-scaled 2.78B: all three trials finished; older recipe and validation. exp262 NoPE + smear: final loss 3.00207 versus its control's 2.97454. Its early advantage reversed, illustrating why interim rankings can mislead. exp157 learned RoPE deltas: corrected CE 3.0654 versus vanilla 3.0815. Only exp277 provides the same native + MPNN recipe as the current wider run. Raw historical validation losses cannot be pooled across cache/logging paths.

# training_comparison.png

Full-validation observations through step 46,508, exported October 19, 2026. Exact-step comparison; no interpolation or smoothing. Same logical training recipe, different hardware/gradient accumulation; single seed, incomplete epoch. The right panel zooms into observations after step 25,000.




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$ plot_training_comparison.py
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